

NSTEMI/ACS — Antithrombotics, DAPT & Vasospastic Angina

ANTICOAGULANT COMPARISON SHELF

ENOXAPARIN REFERENCE

UFH EQUIVALENT

BIVALIRUDIN

FONDAPARINUX

FONDAPARINUX ADVANTAGE = LESS BLEEDING ONLY — NOT BETTER EFFICACY

PRASUGREL DOSE ADJUSTMENT 60 mg LOAD → 10 mg/d; REDUCE TO 5 mg/d IF <60 kg OR >75 yr

PRASUGREL CONTRAINDICATION PRIOR STROKE/TIA → USE TICAORELOR

PHARMACIST DISPENS DAPT PRESCRIPTION

CLONIDOGREL PENALTY BENCH 1 IN 3 ARE POOR RESPONDERS

PRASUGREL: NO CYP2C19 DEPENDENCE → CONSISTENT

CORONARY PHARMACY

NEEDS UFH AT PCI — CATHETER THROMBOSIS RISK

75-162 mg/d

12 MONTHS

P2Y<sub>12</sub>

DES THROMBOSIS = PLATELET-MEDIATED → ANTIPLATELET, NOT ANTICOAGULANT

DABIGATRAN — ANTICOAGULANT

WARFARIN — SAME PROBLEM

PREVENTS FIBRIN CLOT — NOT PLATELET PLUG

VASOSPASTIC ANGINA RESEARCH LAB

SPASM SIMULATION CHAMBER

CORONARY ARTERY IN TIGHT SPASM

TRANSIENT ST ELEVATION — RESOLVES SPONTANEOUSLY

NORMAL CORONARIES — NO OCCLUSION TO FIX

BETA BLOCKER soldier TRYING TO ENTER

UNOPPOSED α → WORSENS SPASM

PROSTACYCLIN — PROTECTIVE

BLOCKS PROSTACYCLIN → WHEN CORONARY GOES INTO → ↑ SPASM

CORRECT TREATMENT CCB — FIRST LINE

TROPOIN NEGATIVE

SMOKING = #1 RISK FACTOR



## 1. Enoxaparin vs UFH

### WHAT YOU SEE

Shelf: 'ENOXAPARIN reference ... UFH equivalent'

### WHAT IT ENCODES

Anticoagulation is mandatory in ACS. Enoxaparin (LMWH) is preferred over UFH for medically managed NSTEMI; UFH is preferred when early PCI is planned (titratable, reversible). Both bridge until revascularization.

### BOARD TRAP

Using LMWH in a patient going straight to cath where UFH is easier to monitor with ACT, or in severe renal failure where UFH is safer.

## 2. Bivalirudin

### WHAT YOU SEE

Bottle 'BIVALIRUDIN'

### WHAT IT ENCODES

Bivalirudin is a direct thrombin inhibitor used during PCI, notably in heparin-induced thrombocytopenia where heparins are contraindicated. Lower bleeding than UFH+GPIIb/IIIa in some trials.

### BOARD TRAP

Reaching for heparin in a patient with prior HIT — use bivalirudin or argatroban instead.

## 3. Fondaparinux — less bleeding

### WHAT YOU SEE

Banner 'FONDAPARINUX ADVANTAGE = LESS BLEEDING ONLY — NOT BETTER EFFICACY'

### WHAT IT ENCODES

Fondaparinux (factor Xa inhibitor) gives the lowest bleeding risk for medically managed NSTEMI but carries catheter-thrombosis risk during PCI — adjunctive UFH must be given if the patient goes to cath.

### BOARD TRAP

Using fondaparinux alone during PCI — it causes catheter thrombosis; add UFH at the time of the procedure.

## 4. P2Y12 × 12 months

### WHAT YOU SEE

'P2Y12' figure + '12 MONTHS' tag

### WHAT IT ENCODES

DAPT = aspirin + a P2Y12 inhibitor (clopidogrel, prasugrel, or ticagrelor) for 12 months after ACS regardless of stent type, then aspirin indefinitely. Ticagrelor/prasugrel are preferred over clopidogrel post-PCI.

### BOARD TRAP

Stopping the P2Y12 inhibitor early after ACS — premature DAPT cessation is a major risk for stent thrombosis.

## 5. Aspirin 75–162 mg

### WHAT YOU SEE

'75–162 mg/d' tag

### WHAT IT ENCODES

Aspirin loading 162–325 mg then lifelong maintenance 75–100 mg/day (often quoted 75–162). It irreversibly inhibits COX-1/TXA2 for the platelet's lifespan (~7–10 days).

### BOARD TRAP

High maintenance aspirin doses add bleeding without efficacy benefit — low-dose (81 mg) is standard for chronic therapy.

## 6. Prasugrel dose & contraindication

### WHAT YOU SEE

'PRASUGREL 60 mg load, 10 mg/d; reduce to 5 mg if <60 kg or >75 yr; CONTRAINDICATION — prior stroke/TIA'

### WHAT IT ENCODES

Prasugrel (irreversible P2Y12) is contraindicated with prior stroke/TIA (net harm), and dose-reduced to 5 mg if <60 kg or age >75. Most potent; avoid before coronary anatomy is known.

### BOARD TRAP

Giving prasugrel to a patient with prior CVA/TIA — it is absolutely contraindicated due to intracranial bleeding risk.

## 7. Clopidogrel CYP2C19 — 1 in 3 poor responders

### WHAT YOU SEE

'CLOPIDOGREL PENALTY BENCH — 1 in 3 are poor responders'

### WHAT IT ENCODES

Clopidogrel is a prodrug needing CYP2C19 activation; loss-of-function alleles → poor response and higher stent thrombosis. Ticagrelor/prasugrel are not affected and are preferred post-ACS PCI.

### BOARD TRAP

Assuming clopidogrel is working in a poor-metabolizer — consider ticagrelor/prasugrel after stent thrombosis on clopidogrel.

## 8. DES = platelet-mediated → antiplatelet

### WHAT YOU SEE

'DES THROMBOSIS = PLATELET-MEDIATED → ANTIPLATELET, NOT ANTICOAGULANT'

### WHAT IT ENCODES

Stent (DES) thrombosis is a platelet-driven white clot — prevented by DAPT (antiplatelets), not by anticoagulants. This is why P2Y12 + aspirin, not warfarin, protect the stent.

### BOARD TRAP

Adding warfarin to prevent stent thrombosis — arterial stent clot is platelet-mediated and needs antiplatelet therapy.

## 9. Dabigatran — fibrin, not platelet

### WHAT YOU SEE

'DABIGATRAN ANTICOAGULANT — PREVENTS FIBRIN CLOT NOT PLATELET PLUG'

### WHAT IT ENCODES

Anticoagulants (dabigatran DTI, warfarin) prevent fibrin-rich red clots (venous/AF stroke), not the platelet plug of arterial stent thrombosis. Right drug, right clot mechanism.

### BOARD TRAP

Using an oral anticoagulant as monotherapy to protect a coronary stent — it targets fibrin, not platelets.

## 10. Vasospastic angina — spasm resolves

### WHAT YOU SEE

'CORONARY ARTERY IN TIGHT SPASM ... TRANSIENT ST ELEVATION — RESOLVES SPONTANEOUSLY ... NORMAL CORONARIES, TROPONIN NEGATIVE'

### WHAT IT ENCODES

Prinzmetal/vasospastic angina: transient resting ST elevation that resolves spontaneously, normal/non-obstructive coronaries, often negative troponin. Smoking is the #1 risk factor; triggered by ergot, cocaine, hyperventilation.

### BOARD TRAP

Calling transient resting ST elevation with clean coronaries a STEMI — it's vasospasm; treat with CCB/nitrates, not thrombolysis.

## 11. Beta-blocker worsens spasm

### WHAT YOU SEE

'BETA BLOCKER soldier trying to enter — UNOPPOSED  $\alpha$  → WORSENS SPASM'

### WHAT IT ENCODES

In vasospastic angina, non-selective beta-blockade leaves alpha-mediated vasoconstriction unopposed and worsens spasm. Avoid beta-blockers; same logic as cocaine chest pain.

### BOARD TRAP

Giving a beta-blocker for vasospastic/Prinzmetal angina (or cocaine MI) — unopposed alpha worsens coronary spasm.

## 12. CCB first-line; avoid ASA/NSAID

### WHAT YOU SEE

'CORRECT TREATMENT — CCB = FIRST LINE; NSAID/ASA blocks PROSTACYCLIN (protective)'

### WHAT IT ENCODES

Treat vasospasm with calcium channel blockers (first-line)  $\pm$  nitrates. High-dose aspirin/NSAIDs inhibit protective prostacyclin (PGI<sub>2</sub>) and can worsen spasm. Smoking cessation is key.

### BOARD TRAP

High-dose aspirin in vasospastic angina — it blocks vasoprotective prostacyclin; CCBs are the cornerstone therapy.